

Practical Exam - Supervised

Step - 1 : Theory

(1) Why is Accuracy a terrible metric for fraud detection?

→ Accuracy is not a good metric for fraud detection because the dataset is highly imbalanced. If 99.83% of transactions are legitimate and only 0.17% are fraudulent, a model that predicts every transaction as "Not fraud" will still achieve 99.83% accuracy, but it will fail to detect any fraud.

Accuracy : 99.83%

Precision (fraud) : 0

Recall (fraud) : 0

(2) What is the precision - Recall tradeoff?

→ Precision and Recall have an inverse relationship. Increasing Recall often decreases Precision, while increasing precision may reduce Recall.

Actual / Predicted	legitimate (0)	fraud (1)
legitimate (0)	TN	FP
fraud (1)	FN	TP

Lowering the threshold (0.5 → 0.2)

- More transactions are classified as fraud.
- Recall increase because more fraud class are detected.
- Precision may decrease because more legitimate transaction are incorrectly flagged as fraud.

(3) Explain three strategies to handle extreme class imbalance.

(a) Random Under sampling

- Removes samples from the majority class.
- Makes the dataset more balanced.
- Faster training but may lose useful information.

(b) SMOTE

- Creates synthetic ex of the minority class.
- Improves fraud detection without removing data.
- Helps the model learn fraud patterns better.

(c) Class Weight = "balanced"

- Assigns higher importance to the minority class during training.
- No data is added or removed.
- Useful for models like Logistic Regression and Random Forest.

(4) What is Precision - Recall AUC (PR - AUC)?

→ PR-AUC is the AU the precision-Recall curve. it measures how well a model balances Precision and Recall across different classification thresholds.

PR-AUC is preferred over ROC-AUC for fraud detection because fraud datasets are highly imbalanced.

(5) Which is more costly : FP or FN ?

→ In fraud detection, a false Negative is usually more costly than a false Positive.

False Positive: A genuine transaction is incorrectly flagged as fraud. The customer may need to verify the transaction, causing temporary inconvenience.

False Negative: A fraudulent transaction is classified as legitimate. This can result in direct financial loss, customer dissatisfaction, and reputational damage to the bank or payment company.